TAOLUE KURT YANG

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RESEARCH INTERESTS

Efficient and scalable AI and data systems: accelerating AI training and inference, and scientific data processing, through compression and quantization. Topics include LLM quantization, KV-cache compression, and speculative decoding; GPU-accelerated data compression; compressed-domain computing; and high-performance computing.

EDUCATION

Temple University, Philadelphia, PA — 2023 – May 2027 (expected)
Ph.D. in Computer Science
Advisor: Prof. Sian Jin

Columbia University, New York, NY — 2022 – 2023
M.S. in Electrical Engineering

University of Glasgow, Glasgow, UK — 2018 – 2022
B.Eng. in Electrical Engineering, First Class Honours

RESEARCH EXPERIENCE

Graduate Research Assistant — 2023 – Present
Temple University, Advisor: Prof. Sian Jin — *Philadelphia, PA*

- **GPU-accelerated genomic data compression:** designed CuVert-Q, GPUFASTQLZ, and GFAz, GPU compressors for sequencing reads and pangenome graphs that resolve the throughput–ratio tradeoff; built the SeqBench benchmark suite. Released CuVert-Q, GFAz, GPUFASTQLZ, and SeqBench as open-source software.
- **Computing over compressed data:** extending this line to pangenome analytics computed directly on grammar-compressed graphs, without decompression (under review at FAST'27).
- **Computation sharing for genomic deep learning:** designed exact per-layer computation sharing for sequence models across thousands of haplotypes, 7.6× faster than window deduplication and 584× faster than standard inference on 1000 Genomes, with identical outputs.
- **LLM-aware lossy compression:** co-developed KVComp, PackKV, and EmbdC, error-bounded compressors that shrink KV-cache and embedding memory for LLM inference while preserving accuracy.
- **Scientific data compression:** contributed to TOPIQ (SC'26), 3D Gaussian Splatting particle compression (SC'26), error-bounded point cloud compression (VLDB'26), cross-field prediction (HPDC'25), and performance modeling for error-bounded lossy compression of HPC simulation data.
- **Interdisciplinary collaboration:** work closely with Prof. Xinghua (Mindy) Shi (Temple, bioinformatics) on biological data compression (CuVert-Q, GFAz, SeqBench, DCC'26, FAST'27).

PUBLICATIONS

Peer-Reviewed Conference Papers

[C1] **Taolue Yang**, Y. Liu, B. Jiang, C. Li, X. Shi, S. Jin. "CuVert-Q: Resolving the Throughput-Ratio Tradeoff in Genomic Sequence Data Compression via GPU Acceleration." *17th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB)*, 2026. **Best Paper Award.**

[C2] **Taolue Yang**, Y. Liu, B. Jiang, X. Shi, S. Jin. "GFAz: State-of-the-Art Graphical Fragment Assembly Compression." *40th ACM International Conference on Supercomputing (ICS)*, pp. 650–661, 2026.

[C3] **Taolue Yang**, Y. Liu, C. Li, X. Shi, S. Jin. "SeqBench: A Benchmark Suite for Lossless and Lossy Compression of Sequence Data." *15th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB)*, 2024.

[C4] Y. Liu, B. Jiang, **Taolue Yang**, S. Di, R. Underwood, S. Jin. "TOPIQ: Statistical Error Propagation for Quantity-of-Interest Prediction under Lossy Compression." *International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, 2026.

[C5] B. Jiang, Y. Liu, **Taolue Yang**, S. Di, S. Jin. "3D Gaussian Splatting for Scientific Particle Data Compression and Rendering." *International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, 2026.

[C6] Y. Liu, L. Zhang, R. Li, B. Jiang, **Taolue Yang**, K. Zhao, S. Di, E. Dragut, S. Jin. "Error-Bounded Point Cloud Compression Using Truncated Octahedron Quantization." *Proceedings of the VLDB Endowment (PVLDB)*, 19(11): 3105–3118, 2026.

[C7] B. Jiang, **Taolue Yang**, Y. Liu, X. He, S. Di, S. Jin. "PackKV: Reducing KV Cache Memory Footprint through LLM-Aware Lossy Compression." *IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, 2026.

[C8] B. Jiang, **Taolue Yang**, Y. Liu, X. He, S. Di, S. Jin. "EmbdC: Error-Bounded Lossy Video Embedding Compression for On-Device LLM Inference." *26th IEEE International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, 2026.

[C9] Y. Liu, W. Jia, **Taolue Yang**, B. Jiang, M. Yin, S. Jin. "Advancing Scientific Data Compression via Cross-Field Prediction." *34th International Symposium on High-Performance Parallel and Distributed Computing (HPDC)*, 2025.

Peer-Reviewed Workshop Papers and Short Papers

[W1] **Taolue Yang**, Y. Liu, B. Jiang, C. Li, X. Shi, S. Jin. "GPUFast-Q: Novel High-Throughput Sequence Data Compression on GPUs Using Fine-Grained Parallelism." *Data Compression Conference (DCC)*, p. 479, 2026.

[W2] **Taolue Yang**, Y. Liu, B. Jiang, S. Jin. "GPUFASTQLZ: An Ultra Fast Compression Methodology for Fastq Sequence Data on GPUs." *SC24 Workshops (SC-W)*, 2024.

[W3] Y. Liu, **Taolue Yang**, S. Jin. "Accurate Performance Modeling and Uncertainty Analysis of Lossy Compression in Scientific Applications." *Data Compression Conference (DCC)*, p. 390, 2025.

[W4] Y. Liu, W. Jia, **Taolue Yang**, M. Yin, S. Jin. "Enhancing Lossy Compression through Cross-Field Information for Scientific Applications." *SC24 Workshops (SC-W)*, 2024.

Under Review

[U1] **Taolue Yang**, Y. Liu, B. Jiang, X. Shi, S. Jin. "Computing over Compression: Pangenome Analytics Directly on Grammar-Compressed Graphs." Submitted to *USENIX Conference on File and Storage Technologies (FAST)*, 2027.

In Preparation

[M1] **Taolue Yang**, S. Jin. "Exact Per-Layer Computation Sharing for Convolutional Sequence Models Across Haplotypes." In preparation, 2026.

Preprint

[P1] B. Jiang, **Taolue Yang**, Y. Liu, C. Zhang, X. He, S. Jin. "KVComp: A High-Performance, LLM-Aware, Lossy Compression Framework for KV Cache." *arXiv:2509.00579*, 2025.

Patents

[T1] Relaxed speculative decoding for accelerated LLM token generation. Oracle Corporation, patent application filed, 2025.

HONORS AND AWARDS

Best Paper Award, ACM-BCB 2026 (for CuVert-Q) 2026
First Class Honours, University of Glasgow 2022

TEACHING EXPERIENCE

Teaching Assistant, Temple University 2024

- **CIS 1068: Program Design and Abstraction** – Spring 2024 (2 sections), Fall 2024 (1 section)
  First required programming course in the CS major: Java, object-oriented programming, searching and sorting.
  Duties: led weekly lab sections, taught algorithm topics (e.g., searching and sorting) and Java programming in lab, graded assignments and exams, and held weekly office hours.
- **CIS 2109: Database Management Systems** – Fall 2024 (1 section)
  Undergraduate data modeling, SQL, and database application development.
  Duties: led weekly lab sections on SQL and database design, graded assignments, and held weekly office hours.

Teaching Interests
Undergraduate: introductory programming and data structures; computer systems and architecture; parallel programming; database systems.
Graduate: LLM systems (efficient training and inference, quantization, KV-cache optimization, serving); high-performance computing and GPU/CUDA programming; data compression for scientific and AI workloads.

INDUSTRY RESEARCH EXPERIENCE

Software Engineer Intern May 2026 – Aug 2026
Google, Cloud AI (Vertex AI) *Sunnyvale, CA*

- Built a large-scale post-training quantization (PTQ) evaluation framework on GCP Slurm that benchmarks quantized LLMs for accuracy, KL divergence from the full-precision model, and serving throughput.
- Established hardware-aware PTQ deployment best practices for Vertex AI, benchmarking and tailoring quantization recipes across GPU platforms (A3 High, A3 Mega, A3 Ultra, A4) and model families.
- Accelerated quantization-aware training (QAT) with GRPO in **NeMo-RL/Megatron** by **70%** by caching and reusing fake-quantized weights; change accepted upstream by NVIDIA.
- Diagnosed and fixed a Megatron weight-folding bug in **NVIDIA ModelOpt**; fix adopted upstream by NVIDIA.
- Researched quantization-aware reinforcement learning for LLM post-training, studying quantization across the training, rollout, and KV-cache stages to reduce compute cost while preserving reasoning quality.

Software Engineer Intern Jul 2025 – Oct 2025
Google, AI and Infrastructure (MaxText) *Kirkland, WA*

- Optimized large-scale Mixture-of-Experts (MoE) LLM inference throughput on TPUs.
- Implemented Mixture of Block Attention (MoBA) in JAX for both training and inference.
- Designed and integrated interleaved prefill and batch decode with MaxText as the JAX inference backend.
- Developed a fully compatible OpenAI-style API server for MaxText, enabling seamless deployment.
- Integrated **lm-evaluation-harness** and **Evalchemy** into MaxText for up to **30**× faster evaluation.

Software Engineer Intern Jan 2025 – Jul 2025
Oracle, MySQL HeatWave *Redwood City, CA*

- Developed an optimized relaxed speculative decoding algorithm that accelerates LLM token generation in Oracle HeatWave clusters by **1.5**×; filed a patent on this work.
- Built an 8-bit quantized LLM inference stack for AMD CPUs, combining grouped weight quantization with SIMD AVX-512 and AMD BLIS int8 GEMM kernels, achieving a **4.5**× speedup over open-source llama.cpp on CPU prefill.
- Developed a grouped KV-cache layout that packs 64 key activations per group for efficient 8-bit storage and AVX-512 GEMM, accelerating long-context prefill by up to **20%** in llama.cpp.
- Optimized weight quantization strategies for the LLM decoding phase using AVX-512, improving CPU inference throughput by **15%** over the Q4_K_M baseline.

AI Engineer Intern Jun 2023 – Sep 2023
Oortech *New York, NY*

- Developed a LangChain agent for an LLM-based chatbot, enabling semantic search over user queries.
- Implemented a Python web crawler to collect data for LLM fine-tuning and chatbot development.
- Built a text splitter and Python executor for the data loader, supporting structured inputs to the agent.

OPEN-SOURCE CONTRIBUTIONS

NVIDIA NeMo-RL (upstream contributions)

- #3226: FP8 KV-cache scale propagation in the colocated refit path.
- #3441: Avoiding redundant weight fake-quantization during QAT logprob computation.
- #3556: Caching fake-quantized weights across QAT training microbatches.

NVIDIA ModelOpt

- #2131: Debugging `fold_weight` behavior with Megatron tied embeddings.

Research software: CuVert-Q, GFAz, GPUFASTQLZ (GPU genomic compressors); SeqBench (benchmark suite).
Other projects: MaxText, vLLM, lm-evaluation-harness.

ACADEMIC SERVICE

Conference Reviewer (7 papers reviewed)

- ACDSA 2026 and 2027 (5 papers): Intl. Conf. on AI, Computer, Data Sciences, and Applications
- ICECET 2026 (2 papers): Intl. Conf. on Electrical, Computer and Energy Technologies

TECHNICAL SKILLS

Languages: C++, CUDA, Python **Frameworks:** PyTorch, JAX, Megatron, vLLM, SGLang